

Lecture 3: Bond Price Volatility, Duration, Convexity, and Bond Portfolio Strategies

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Learning Objectives

By the end of this lecture you should be able to:

- Review and interpret the price–yield relationship for option-free bonds.
- Explain the main price-volatility properties of option-free bonds.
- Identify the bond characteristics that affect price volatility.
- Compute and interpret price value of a basis point, yield value of a price change, duration, and convexity.
- Use duration and convexity to approximate percentage and dollar price changes.
- Distinguish between analytical duration and empirical duration.
- Explain spread duration, portfolio duration, key rate duration, and yield curve reshaping duration.
- Understand practical limitations of duration as a risk measure.
- Explain why duration is not simply a measure of time.
- Interpret the duration of inverse floaters.
- Describe how bond portfolio responsiveness changes under nonparallel shifts in the yield curve.
- Explain the main bond portfolio strategy frameworks, including benchmark-based, absolute return, and liability-driven strategies.
- Understand the role of bond benchmarks and the limitations of market-cap-weighted bond indexes.
- Distinguish between top-down and bottom-up portfolio construction.
- Explain active strategies, smart beta approaches, and the use of leverage in bond portfolio management.

1 Review of the Price–Yield Relationship for Option-Free Bonds

For an option-free bond, price and yield move in opposite directions. When the required yield rises, the bond price falls. When the required yield falls, the bond price rises. This

inverse relationship follows directly from the bond pricing formula because future cash flows are discounted at the required yield.

For a bond with coupon payment C , face value F , maturity n , and required yield y , the price is

$$P = \sum_{t=1}^n \frac{C}{(1+y)^t} + \frac{F}{(1+y)^n}.$$

Because the denominator increases when y increases, the present value of each cash flow falls, and therefore the bond price falls.

i Worked Example

Review of the Price–Yield Relationship

Suppose a 3-year bond has:

- Face value = \$1,000
- Annual coupon = \$60
- Required yield = 5%

Its price is

$$P = \frac{60}{1.05} + \frac{60}{1.05^2} + \frac{1060}{1.05^3} = 57.14 + 54.42 + 915.91 = 1027.47.$$

If the required yield rises to 7%, the new price is

$$P = \frac{60}{1.07} + \frac{60}{1.07^2} + \frac{1060}{1.07^3} = 56.07 + 52.40 + 865.38 = 973.85.$$

The bond price falls from \$1,027.47 to \$973.85 when the required yield rises from 5% to 7%.

2 Price-Volatility Characteristics of Option-Free Bonds

The price-volatility properties of option-free bonds describe how different bonds respond to changes in required yield.

2.1 Property 1

Although the prices of all option-free bonds move in the opposite direction from the change in required yield, the **percentage price change is not the same for all bonds**.

Some bonds are more sensitive to changes in yield than others. Longer maturity bonds and lower coupon bonds tend to experience greater percentage price changes for a given change in yield.

2.2 Property 2

For **very small changes** in the required yield, the percentage price change for a given bond is roughly the same whether the required yield increases or decreases.

This is the local linear approximation behind duration.

2.3 Property 3

For **large changes** in the required yield, the percentage price change is not the same for an increase in the required yield as it is for a decrease in the required yield.

This occurs because the price–yield relationship is curved rather than linear.

2.4 Property 4

For a given large change in basis points, the **percentage price increase is greater than the percentage price decrease**.

This is the practical consequence of convexity. The curvature of the price–yield relationship makes price gains from falling yields larger than price losses from equally sized yield increases.

2.5 Characteristics of a Bond That Affect Its Price Volatility

The main bond characteristics affecting price volatility are:

- Time to maturity
- Coupon rate
- Yield to maturity
- Embedded options

For option-free bonds:

- Longer maturity generally means greater price volatility.
- Lower coupon generally means greater price volatility.
- Lower yield generally implies greater price volatility.

2.6 Effects of Yield to Maturity

For two otherwise identical option-free bonds, the bond with the lower yield to maturity will exhibit greater percentage price volatility for a given change in yield. This effect is related to the curvature of the present value function.

3 Measures of Bond Price Volatility

Several measures are used to summarize how sensitive a bond's price is to changes in yield.

3.1 Price Value of a Basis Point

The **price value of a basis point (PVBP)** measures the change in the price of a bond for a one-basis-point change in yield.

A basis point is

$$0.01\% = 0.0001.$$

PVBP is often approximated by repricing the bond at yields y and $y + 0.0001$ and taking the difference.

$$PVBP \approx P(y) - P(y + 0.0001).$$

It is a dollar measure of interest-rate sensitivity.

3.2 Yield Value of a Price Change

The **yield value of a price change (YVPC)** is the change in yield associated with a specified change in price. It is the inverse way of thinking about price sensitivity.

3.3 Duration

Duration is one of the most important measures of bond price sensitivity.

3.3.1 Macaulay Duration

The **Macaulay duration** is the weighted average time to receipt of the bond's cash flows, where the weights are the present values of the cash flows as a proportion of the bond price.

$$D_M = \frac{\sum_{t=1}^n t \cdot \frac{CF_t}{(1+y)^t}}{P}$$

For bonds with semiannual coupons, duration is first computed in periods and then divided by the number of periods per year.

3.3.2 Modified Duration

The **modified duration** converts Macaulay duration into a direct measure of price sensitivity:

$$D^* = \frac{D_M}{1+y}$$

for annual compounding, or more generally

$$D^* = \frac{D_M}{1+y/m}$$

when payments occur m times per year.

Modified duration gives the approximate percentage price change for a small change in yield:

$$\frac{\Delta P}{P} \approx -D^* \Delta y$$

3.3.3 Derivation of the Duration Approximation

Start with the bond price formula:

$$P(y) = \sum_{t=1}^n \frac{CF_t}{(1+y)^t}$$

Differentiate with respect to y :

$$\frac{dP}{dy} = \sum_{t=1}^n -t \frac{CF_t}{(1+y)^{t+1}}$$

Divide by P :

$$\frac{1}{P} \frac{dP}{dy} = -\frac{1}{1+y} \cdot \frac{\sum_{t=1}^n t \frac{CF_t}{(1+y)^t}}{P}$$

This yields

$$\frac{1}{P} \frac{dP}{dy} = -D^*$$

which leads to the approximation

$$\frac{\Delta P}{P} \approx -D^* \Delta y.$$

3.4 Properties of Duration

For option-free bonds:

- Duration is less than maturity for coupon bonds.
- Zero-coupon bond duration equals its maturity.
- Duration increases with maturity, but at a decreasing rate.
- Duration decreases as coupon rate increases.
- Duration decreases as yield increases.

3.5 Approximating the Percentage Price Change

Modified duration gives the first-order approximation:

$$\frac{\Delta P}{P} \approx -D^* \Delta y$$

3.6 Approximating the Dollar Price Change

Multiply the percentage change by the bond price:

$$\Delta P \approx -D^* P \Delta y$$

3.7 Spread Duration

Spread duration measures the sensitivity of a bond's price to a change in its credit spread, holding the benchmark yield curve fixed.

It is especially important for corporate bonds, high-yield bonds, and structured products.

3.8 Portfolio Duration

Portfolio duration is the weighted average duration of the holdings when weights are based on market value:

$$D_P = \sum_{i=1}^N w_i D_i$$

where

$$w_i = \frac{\text{Market Value of Bond } i}{\text{Total Portfolio Market Value}}.$$

3.9 Analytical versus Empirical Duration

- **Analytical duration** is computed from a pricing formula.
- **Empirical duration** is estimated by repricing the bond under shifted yields and observing the resulting price change.

Empirical duration is often necessary for bonds with uncertain cash flows or embedded options.

4 Convexity

Duration gives a linear approximation, but the price–yield relationship is curved. Convexity captures that curvature.

4.1 Intuition and Formula

Convexity measures the rate at which duration itself changes as yield changes. It is the second derivative effect in the price–yield relationship.

A common convexity measure for an option-free bond is

$$\text{Convexity} = \frac{1}{P} \sum_{t=1}^n \frac{CF_t t(t+1)}{(1+y)^{t+2}}$$

for annual periods, with adjustments as needed for more frequent compounding.

4.2 Measuring Convexity

Convexity can also be estimated empirically by repricing the bond at yields above and below the current yield.

4.3 Approximating Percentage Price Change Using Duration and Convexity

A better approximation than duration alone is

$$\frac{\Delta P}{P} \approx -D^* \Delta y + \frac{1}{2} \text{Convexity} (\Delta y)^2$$

4.4 Value of Convexity

Convexity improves approximation accuracy for large changes in yield. It also explains why price gains from falling yields exceed price losses from equal yield increases.

4.5 Properties of Convexity

For option-free bonds:

- Convexity is positive.
 - Convexity increases with maturity.
 - Convexity tends to be greater for lower coupon bonds.
 - Convexity tends to be greater at lower yields.
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5 Additional Concerns When Using Duration

Duration is useful, but it has limitations:

- It is a local approximation and works best for small yield changes.
 - It assumes parallel shifts in the yield curve unless modified.
 - For bonds with embedded options, duration can change sharply when yields change.
 - It does not capture curvature unless convexity is added.
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6 Do Not Think of Duration as a Measure of Time

Although Macaulay duration has units of time, duration should not be interpreted simply as “time to maturity” or “average life” in a casual sense.

For risk management purposes, duration is best understood as a **measure of price sensitivity to yield changes**, not merely as a measure of time.

7 Approximating a Bond's Duration and Convexity Measure

In practice, duration and convexity are often approximated numerically by repricing the bond at yields above and below the current yield.

An empirical approximation to duration is

$$D^* \approx \frac{P_- - P_+}{2P_0\Delta y}$$

where:

- P_- = price if yield decreases
- P_+ = price if yield increases
- P_0 = current price

An empirical approximation to convexity is

$$\text{Convexity} \approx \frac{P_- + P_+ - 2P_0}{P_0(\Delta y)^2}$$

7.1 Duration of an Inverse Floater

The duration of an inverse floater can be very large because its coupon moves inversely with rates. The bond's cash flows become more favorable when rates fall and less favorable when rates rise, increasing price sensitivity.

7.2 Key Rate Duration

Key rate duration measures the sensitivity of a bond's price to changes in a specific maturity point on the yield curve, holding other points constant.

This is essential when yield curve shifts are not parallel.

8 Measuring a Bond Portfolio's Responsiveness to Nonparallel Changes in Interest Rates

When the yield curve changes in a nonparallel way, standard duration is not enough. A portfolio can be exposed differently to short-, intermediate-, and long-term rates.

8.1 Yield Curve Reshaping Duration

Yield curve reshaping duration measures the responsiveness of a portfolio to twists, steepenings, flattenings, and other nonparallel curve movements.

This is especially important for active bond portfolio management.

9 The Asset Allocation Decision

Bond portfolio management begins with the asset allocation decision.

9.1 How Much Should Be Allocated to Bonds?

The allocation to bonds depends on:

- Risk tolerance
- Investment horizon
- Liability structure
- Return objectives
- Regulatory and institutional constraints

9.2 Who Should Manage the Bond Portfolio?

Management may be internal or external, depending on institutional size, expertise, and governance structure.

10 Portfolio Management Team

A bond portfolio management team may include:

- Chief investment officer
- Portfolio managers
- Credit analysts
- Quantitative analysts
- Traders

- Risk managers

Effective bond portfolio management requires both macro views and security-level analysis.

11 Spectrum of Bond Portfolio Strategies

Bond strategies span a wide spectrum.

11.1 Bond Benchmark-Based Strategies

These strategies seek to outperform or closely track a benchmark index.

11.2 Absolute Return Strategies

These strategies focus on achieving a target return regardless of benchmark composition.

11.3 Liability-Driven Strategies

These strategies are designed to match assets to the behavior of liabilities, especially for pension funds and insurance companies.

12 Bond Benchmarks

Bond benchmarks play a central role in performance measurement and portfolio construction.

12.1 Bond Market Indexes

Bond market indexes summarize the performance of broad segments of the bond market.

12.2 Market-Capitalization-Weighted Bond Indexes

These indexes weight securities by the amount of debt outstanding.

12.3 Commonly Used Bond Market Indexes

Examples include major government, corporate, aggregate, and global bond indexes.

12.4 Problems Associated with Market-Cap-Weighted Bond Indexes

A major criticism is that the largest borrowers receive the largest weights, which may not be desirable from a risk-adjusted perspective.

12.5 Customized Bond Indexes

Institutions often build custom indexes tailored to liabilities, risk limits, or mandate constraints.

12.6 Alternative Bond Indexes

Alternative weighting methods seek to avoid weaknesses of market-cap weighting.

12.7 Ad Hoc Weighted Index

Some indexes use judgment-based or rule-based weighting schemes.

12.8 Risk-Based Schemes

Risk-based weighting allocates more deliberately with respect to duration, volatility, spread, or other risk factors.

13 The Primary Risk Factors

Important bond portfolio risk factors include:

- Level of interest rates
 - Shape of the yield curve
 - Credit spreads
 - Liquidity
 - Inflation
 - Currency risk for global portfolios
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14 Top-Down versus Bottom-Up Portfolio Construction and Management

Bond portfolios may be built using different philosophies.

14.1 Manager Expectations versus the Market Consensus

Active management depends on views that differ from what is already reflected in market prices.

14.2 Interest-Rate Expectations Strategies

Managers may position portfolios based on expected changes in the level of rates.

14.3 Yield Curve Strategies

Managers may position portfolios for steepening, flattening, or curvature changes in the yield curve.

15 Active Portfolio Strategies

Active strategies attempt to add value through:

- Duration positioning
 - Yield curve positioning
 - Sector rotation
 - Credit selection
 - Security selection
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16 Smart Beta Bond Strategies

Smart beta bond strategies apply systematic, rule-based approaches that deviate from traditional market-cap weighting. They may tilt toward value, carry, quality, low volatility, or other characteristics.

17 The Use of Leverage

Leverage can increase exposure to return opportunities, but it also magnifies risk. In bond portfolios, leverage may be obtained through borrowing, derivatives, or repo transactions.

Key Points

- Bond prices move inversely with yields, and the price–yield relationship for option-free bonds is convex.
- Different option-free bonds do not have the same price volatility. Maturity, coupon rate, and yield level all affect sensitivity to changes in yield.
- Duration is the primary first-order measure of bond price sensitivity, while convexity captures second-order effects.
- Duration provides a good approximation for small yield changes; duration plus convexity provides a better approximation for larger changes.

- Duration should be interpreted mainly as a measure of price sensitivity, not merely as a measure of time.
- Key rate duration and yield curve reshaping duration are needed when yield curve shifts are not parallel.
- Spread duration is important for spread products such as corporate bonds and structured products.
- Bond portfolio management involves asset allocation, benchmark choice, strategy selection, and risk management.
- Benchmark-based, absolute return, and liability-driven approaches represent different portfolio management frameworks.
- Bond indexes are useful but imperfect; market-cap-weighted indexes have important limitations.
- Active strategies require views that differ from market consensus.
- Smart beta and leveraged strategies extend the range of bond portfolio approaches.